

His10-MBP-Cas9 purification

Lysis Buffer – 200 mL:

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| • 50 mM Tris-Cl, pH 7.5 | 10 mL 1M |
| • 500 mM NaCl | 25 mL 4M |
| • 1 mM TCEP | 400 µL 0.5M |
| • 5% glycerol | 10 mL 100% |
| • 0.5mM PMSF | 250 µL 400 mM |
| • protease inhibitors (Roche) | 4 tablets |

Wash Buffer – 500 mL:

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| • 50 mM Tris-Cl, pH 7.5 | 25 mL 1M |
| • 500 mM NaCl | 62.5 mL 4M |
| • 1 mM TCEP | 1 mL 0.5M |
| • 10 mM imidazole | 1.67 mL 3M |
| • 5% glycerol | 25 mL 100% |

Elution Buffer – 200 mL:

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| • 50 mM Tris-Cl, pH 7.5 | 10 mL 1M |
| • 500 mM NaCl | 25 mL 4M |
| • 1 mM TCEP | 0.4 mL 0.5M |
| • 300 mM imidazole | 20 mL 3M |
| • 5% glycerol | 10 mL 100% |

1. Harvest cells from large-scale overnight cultures by spinning down in 1 L centrifuge bottles: 20 minutes at 4000 rpm, 4 °C
 - a. NOTE: one of the two centrifuges can only go up to max speed of 3400 rpm
2. Pour off supernatant and either freeze cell pellet (in centrifuge bottle) at -80 °C until purification, or proceed to step #3. (Better to not keep pellets in centrifuge bottles too long, since this will deplete the lab supply.
3. Resuspend cell pellets in lysis buffer. I use 10-15 mL per 1 L of culture. Multiply accordingly based on the number of flasks you grew. Either use resuspended cells and proceed directly to step #4, or transfer to 50 mL Falcon tubes, flash freeze in liquid nitrogen, and store at -80 °C until purification.

Save 20 µL for SDS-PAGE analysis: "cell pellet"

4. Lyse cells by sonication. Do not do this in glass containers (they can shatter). Either sonicate directly in 50 mL Falcon tubes, or Oakridge tubes, or for large volumes, in plastic beakers. For small cultures I use the sharp tipped needle on the sonicator, but usually (for resuspended pellets deriving from >2 L of culture) I use the blunt-ended tip on the sonicator. Make sure it is screwed on tightly. Before setting up sonication, wash the tip in H₂O and EtOH, and dry before use with Kimwipe.
 - a. Sonicate at power setting 6.0: 10 seconds on, 20 seconds off, total run time of 5 minutes. Cells must be on ice during this step, as the temperature will increase during sonication and the cells must stay cold.

5. Clarify cell lysate by centrifugation at 15,000 rpm for 30 minutes, 4 °C. *It is critical that tubes are precisely balanced for these rotors spinning at these high speeds.* Use the digital balance and balance to within +/- 0.01 g.
6. During the 30 minute spin in step #5, equilibrate Ni-NTA resin in wash buffer. I generally use a column volume of ~5 mL resin for 2-4 L of culture, and then scale accordingly if growing less or more. (Note that the Ni-NTA stock bottles are a 50% (v/v) suspension in EtOH, and so for 5 mL column volume you need to use 10 mL of this suspension.
 - a. To equilibrate, transfer resin into 50 mL Falcon tube, and add mQ H₂O to about the 50mL mark. Invert a few times to fully resuspend the resin. Centrifuge at 2400 rpm (*no higher!*) for 3 minutes, and gently pour off supernatant into the sink immediately after spin is over. These pellets are quite fragile, and will begin to dissipate in the supernatant rather quickly. Avoid accidentally pouring off your Ni-NTA resin.
 - b. Repeat #7a by adding more mQ H₂O, spinning, and pouring off supernatant.
 - c. Repeat #7b 2-3 times, but using wash buffer instead.
7. After the spin in step #5 concludes, **save 20 µL for SDS-PAGE analysis: "cell lysate"**
 - a. Then, transfer entire supernatant (by decanting or with a serological pipette... I prefer a pipette) to the Falcon tube containing your Ni-NTA resin equilibrated in wash buffer. If volume of supernatant is larger than will fit into one Falcon tube, you can split sample into 2 (or more), being sure that the Ni-NTA resin and your lysate is evenly mixed between each. Be careful to avoid disturbing the cell debris pellet when removing the supernatant.
8. Rock the Ni-NTA resin with cell lysate at 4 °C, 30-60 minutes, in the Falcon tube.
9. Pour Ni-NTA resin with lysate into disposable plastic (or glass) column. If using a glass column, be sure to thoroughly wash with copious amounts of EtOH and water before use.
Save 20 µL of the flow-through for SDS-PAGE analysis: "flow-through, FT." Aside from this, the flow-through will not be saved and so can be collected in a waste container.
10. Wash the resin with 5 column volumes.
Save 20 µL of the flow-through for SDS-PAGE analysis: "wash #1, W1." Aside from this, the washes will not be saved and can be collected in a waste container.
11. Repeat step #10 4 more times, for a total of 5 washes.
Save 20 µL of the flow-through for SDS-PAGE analysis: "W2, W3, etc." Aside from this, the washes will not be saved and can be collected in a waste container.
12. Elute by adding 1 column volume of elution buffer, 5 times. *Be sure to collect these fractions which contain our protein of interest!* Collect each elution fraction in a separate 15 mL Falcon tube.
Save 20 µL of each elution fraction for SDS-PAGE analysis: "elution #1, E1, E2, etc."
13. During steps #9-12, also monitor the amount of protein in each fraction via the Bradford assay.
 - a. Bradford reagent undergoes a color change in the presence of protein. It can be used to do quantitative protein concentration measurements, or in this case, give us a quick colorimetric read-out for: if our washes succeeded then there should be no protein coming off the column during wash #5, and so no color change. Similarly, Bradford assay will tell us which of our elution fractions contain a substantial amount of our protein of interest (or any protein at all!).

- b. Note that one could also use the $A_{280\text{nm}}$ reading to determine protein concentration, but it's more work at this stage then necessary.
14. Based on Bradford, pool the elution fractions that contain protein. Measure the absorbance at 280 nm and record this, along with the volume, to give us the overall yield. Also, print out the absorbance spectrum for your notebook, since this has information on the purity of protein (or, if it is co-purifying with any nucleic acid).
15. If cleaving off the His₁₀-MBP fusion, add TEV protease to elution fractions. The TEV protease purified by Kaihong is typically 0.5 mg per tube (250 μL); use 1 tube for every 10–20 mg of purified protein.
16. After adding TEV protease to the eluate, dialyze overnight in Wash Buffer using a Slidealyzer cassette, MWCO = 10k or 30k. Make sure the buffer is pre-chilled to 4 °C

NEXT DAY

Cas9 IEX Buffer A – 1500 mL:

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| • 20 mM Tris-Cl, pH 7.5 | 30 mL 1M |
| • 125 mM KCl | 93.75 mL 2M |
| • 1 mM TCEP | 3 mL 0.5M |
| • 5% glycerol | 75 mL 100% |

Cas9 IEX Buffer B – 500 mL:

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|-------------------------|------------|
| • 20 mM Tris-Cl, pH 7.5 | 20 mL 1M |
| • 1 M KCl | 250 mL 2M |
| • 1 mM TCEP | 1 mL 0.5M |
| • 5% glycerol | 25 mL 100% |

Cas9 Gel Filtration Buffer – 1000 mL:

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| • 20 mM Tris-Cl, pH 7.5 | 20 mL 1M |
| • 200 mM KCl | 100 mL 2M |
| • 1 mM TCEP | 2 mL 0.5M |
| • 5% glycerol | 50 mL 100% |

17. Run TEV-cleaved Cas9 back over the Ni-NTA resin, being sure to equilibrate the resin in Wash Buffer first (>10 CV). Collect the flow-through. Add 1 column volume of wash buffer and collect with the flow-through.
18. Place TEV-cleaved Cas9 back into a Slidealyzer cassette, and dialyze into IEX Buffer A for 2–4 hours. It is critical that this dialysis not proceed longer, as Cas9 is not very soluble at low salt and can will precipitate with prolonged incubation. (The disadvantage of raising the salt concentration is that Cas9 will no longer bind the SP column efficiently.)
19. Purify Cas9 by ion exchange, using a HiTrap HP SP column. Protocol: 0–100% Buffer B over the course of 20 column volumes. Cas9 should elute at ~26 %B. Concentrate to <2 mL.
20. Purify Cas9 by gel filtration, using a Superdex 200 16/60 column. Concentrate and snap freeze.